Solar Energy Management

for Home Assistant

Intelligent solar optimization, EV charging, battery management, and peak load control — all in one integration with a beautiful real-time dashboard.

Version 2.7 · Home Assistant 2025.3+

What is SEM?

Solar Energy Management (SEM) is a comprehensive Home Assistant integration that monitors your solar installation, optimizes EV charging, manages battery storage, tracks costs, and visualizes everything in a stunning real-time dashboard.

Solar Optimization

Dynamically adjusts EV charging current (6-32A) to match real-time solar surplus. Maximizes self-consumption.

Battery Management

4-zone SOC strategy with priority charging, buffer zones, and automatic battery-assist for EV charging.

EV Charging

Solar, night, and manual charging modes. Smart forecast-based night reduction. Session tracking with cost attribution.

Peak Load Control

15-minute rolling peak tracking with automatic load shedding. Drag-and-drop device priority ordering.

Cost Tracking

Daily, monthly, and yearly cost/savings/revenue tracking. Dynamic tariff support. Demand charge monitoring.

Environmental Impact

CO2 avoided and trees-equivalent calculations. Growing icon visualization from sprout to forest.

Supported Hardware

- **Inverters:** Huawei SUN2000, SolarEdge, Fronius, and any inverter with HA integration
- **Batteries:** Huawei LUNA2000, BYD, Tesla Powerwall, and others
- **EV Chargers:** KEBA P30, Easee, go-eCharger, Wallbox, Tesla Wall Connector
- **Grid Meters:** Any smart meter with power and energy sensors in HA

Dashboard

SEM generates a 7-tab dashboard with a unified glassmorphism dark theme. All cards feature dot grid backgrounds, radial gradients, glow effects, and smooth hover transitions.

Tab	Description
Home	Animated system diagram with device nodes, solar summary card with glow ring, 7-day energy chart, weather card, smart recommendations
Energy	Sankey flow diagram, energy distribution donut, self-consumption & autarky gauges, 24h power curves, CO2 avoided, trees saved
Battery	SOC radial gauge, charge/discharge history, SOC zone configuration sliders
EV	Charging status, session energy & cost, solar share gauge, lifetime statistics
Control	Drag-and-drop load priority, EV/battery/solar/tariff settings, observer mode
Costs	Today/month/year KPIs, period-reactive charts, EV economics, demand charge, tariff rates
System	Health diagnostics, sensor availability, configuration summary

Custom Cards

SEM includes 6 custom Lovelace cards built specifically for the dashboard:

- **System Diagram** — animated SVG with hub-spoke layout, glow rings, flow dots, device nodes
- **Solar Summary** — production metrics with animated SVG glow ring and forecast
- **Weather** — live clock, weather conditions, colored temperature forecast bars
- **Chart Card** — Chart.js-powered with 6 presets (costs, savings, energy, power, battery, EV)
- **Period Selector** — date range picker that controls all chart cards
- **Load Priority** — drag-and-drop device ordering with real-time power display

Color Scheme

- Solar (orange)
- Grid Import (steel blue)
- Grid Export (purple)
- Battery Discharge (teal)
- Battery Charge (pink)
- Home Consumption (cyan)
- EV Charging (green)

Sensors & Data

SEM creates 150+ sensors organized by category, all integrated with the HA Energy Dashboard.

Category	Sensors	Examples
Power	9	Solar, grid, battery, EV, home consumption (W)
Daily Energy	7	Solar yield, grid import/export, battery, EV, home (kWh)
Monthly Energy	6	All sources aggregated per month (kWh)
Yearly Energy	7	Year-to-date totals, reset on Jan 1 (kWh)
Energy Flows	9	Solar-to-home, solar-to-EV, grid-to-battery, etc. (W + kWh)
Daily Costs	5	Import costs, savings, export revenue, net cost, battery savings (CHF)
Monthly Costs	4	Monthly aggregated financials (CHF)
Yearly Costs	5	Year-to-date financials (CHF)
Environmental	5	Daily/yearly/lifetime CO2 avoided (kg), trees equivalent
Performance	4	Self-consumption rate, autarky rate, efficiency (%)
Charging	10	State, strategy, current, session energy/cost/solar share
Forecast	8	Today/tomorrow kWh, peak time, surplus window
Load Mgmt	10	Peak tracking, device count, load management status
Tariff	8	Import/export rates, price level, dynamic pricing

Charging Modes

☀️ Solar Charging

Dynamically adjusts EV current to match solar surplus. Pauses when clouds reduce production, resumes automatically.

🌙 Night Charging

Grid charging during off-peak hours (configurable window). Smart forecast reduction lowers target when tomorrow is sunny.

🔋 Battery Assist

When battery SOC exceeds the buffer threshold, battery supplements solar to maintain EV charging through cloudy periods.

👁 Observer Mode

Read-only monitoring — SEM tracks all data but sends no commands to hardware. Safe for testing.

Setup

SEM uses the HA Energy Dashboard as its primary configuration source. It auto-detects solar, grid, battery, and individual device sensors. The 3-step setup wizard asks for EV charger sensors, battery capacity, and peak limit — everything else uses smart defaults.